

Thinking Image Color Aesthetics Assessment

ICCV 2023 – Summary

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Motivation

- **ICAA**: Image Color Aesthetics Assessment
 - Evaluates **color harmony, combinations, perception**
- Different from **IAA**: focuses only on color, not overall aesthetics
- **Challenges**:
 1. Complex color types & combinations, but human perception limited (~8 colors)
 2. Highly subjective color preferences
 3. Lack of large-scale, color-focused datasets & fair benchmarks

Contributions

1. Delegate Transformer

- Deformable attention to select **interest points** for colors
- Color Space Segmentation module mimics human perception

2. ICAA17K Dataset

- 17,726 images, 30 popular color combos, 80 devices, 50 scenes
- ~1,500 human ratings per image
- Balanced via **pre-assessment strategy**

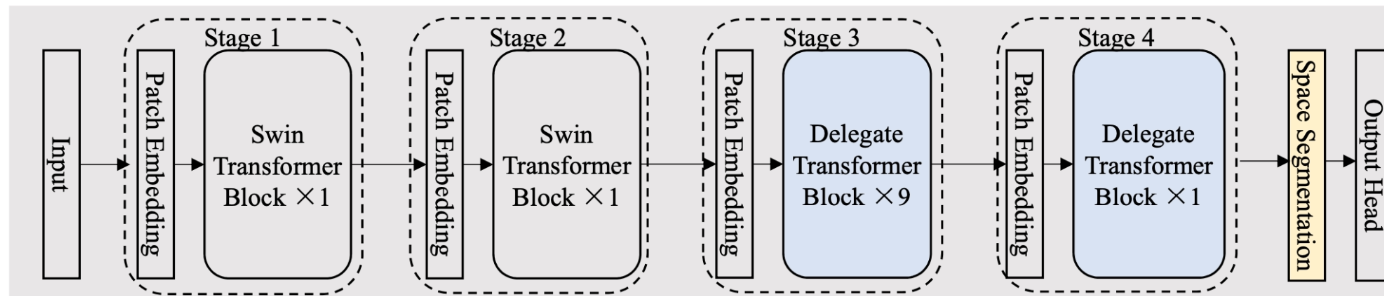
3. Comprehensive Benchmark

- 15 SOTA models tested on ICAA17K & SPAQ

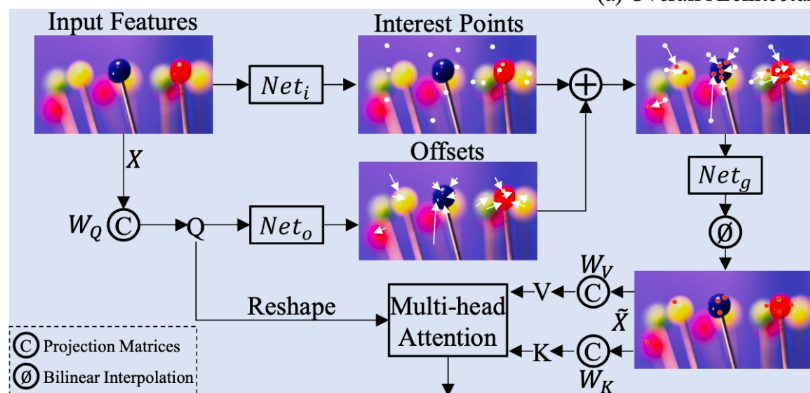
Overall Architecture

- **Stage 1-2: Swin Transformer**
 - Local attention (Shifted Window)
 - Aggregates local aesthetic features
- **Stage 3-4: Delegate Transformer**
 - Sparse attention (Interest Points)
 - Color Space Segmentation module
- **Output Head:**
 - Predicts color aesthetic score
 - Optional multitask: holistic aesthetics score

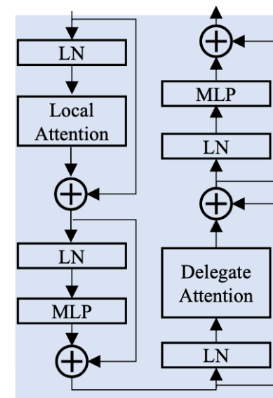
Model Overview



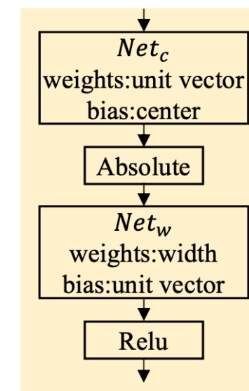
(a) Overall Architecture



(b) Delegate Attention Module



(c) Delegate Transformer Block



(d) Color Space Segmentation Module

Figure 2. The proposed model. (a) Overall architecture. (b) Feature processing flow. The input features are sent to learn the original interest points and adjusted by offsets, then filtered by the gate module. (d) The entire block. (d) The color space segmentation module, which is modeled as a stack of CNN layers.

Delegate Attention – Motivation

Standard Multi-Head Self-Attention (MSA):

$$MSA = \text{Softmax} \left(\frac{Q^{(m)} K^{(m)T}}{\sqrt{D}} \right) V^{(m)T}$$

- D : dimension of Q/K
- m : attention head index

Issue: Computing on **entire feature map** is costly and attention is diffuse.

Delegate Attention – Key Steps

1. Generate Interest Points

$$X_i = \oslash [X; G \cdot (Net_i(X) + Net_o(Q))]$$

- Net_i : predicts initial coordinates of interest points
- Net_o : computes **offsets** to adjust coordinates
- G : gate module to activate/deactivate points

2. Gate Module

$$G = \sigma (Net_g(X) \cdot \alpha)$$

- σ : Sigmoid function
- α : large constant $\rightarrow$ gate outputs close to 0 or 1
- Removes redundancy & prevents overlaps

Delegate Attention – Bilinear Sampling

Bilinear Interpolation for feature extraction:

$$\oslash[X; (p_x, p_y)] = \sum_{(q_x, q_y)} M(p_x, q_x) \cdot M(p_y, q_y) \cdot X(q_x, q_y)$$

Where:

- $M(a, b) = \max(0, 1 - |a - b|)$
- Only 4 nearest neighbors contribute

Color Space Segmentation – Goal

- Group interest points into κ **color spaces**
- Assign different attention weights to **dominant** vs **secondary** colors

Color Space Segmentation – Formula

For interest point p_j in κ -th color space:

$$s_i(p_j) = \max\{0, 1 - w_i \cdot |p_j - c_i|\} \text{ Where:}$$

- $s_i(p_j)$: weight of p_j in i -th color space
- w_i : width of the i -th color space
- c_i : center of the i -th color space

Color Space Segmentation – Learnable Parameters

- **Net_w** learns widths (w_i)
- **Net_c** learns centers (c_i)
- Both are lightweight CNNs → parameters updated via backprop
- Avoids fixed bins; adapts to image-specific color distributions
- **Optimal κ** : 7 (found in experiments)

Advantages of This Design

- **Efficiency:** Sparse attention → ~40% fewer FLOPs
- **Human-like perception:** Focus on dominant color regions
- **Adaptability:** Learnable widths/centers for flexible color grouping
- **Better fine-grained perception:** Matches subjective color importance

ICAA17K Dataset

Design Goals

- Avoid selection bias & long-tail color distribution
- Ensure diversity in colors, scenes, devices

Diversity

- 15 complementary/polychromatic + 15 monochromatic color types
- 50+ scene categories (animals, plants, landscapes, night scenes)
- 80+ devices (phones & cameras)

Pre-Assessment Strategy: Color Harmony Score

$$f_{obj}(C) = \sum_{l \in C} \|H(l) - E_T^{(\beta)}(l)\| \cdot Y(l)$$

Where:

- C : input image
- l : pixel in the image
- $H(l)$: hue of pixel l
- $Y(l)$: saturation of pixel l
- $T(\beta)$: harmony template (e.g., complementary, triadic) with rotation β
- $E_T^{(\beta)}(l)$: boundary hue from the template
- $\|\cdot\|$: arc-length distance on hue wheel (in radians)

Meaning:

Smaller distance $\rightarrow$ closer to harmony template $\rightarrow$ higher objective color quality.

Pre-Assessment Strategy: Step 2 – f_{subj} : Popularity Score

$$f_{\text{subj}}(C) = \frac{FA(C)}{\sqrt{VI(C)}}$$

Where:

- $FA(C)$: number of “faves” (likes) on Flickr
- $VI(C)$: number of “views” on Flickr

Meaning:

- Higher fave/view ratio → potentially more visually appealing
- Square root in denominator → reduce extreme influence of viral images

Pre-Assessment Strategy: Step 3 – Score Combination & Sampling

1. Normalization

- Normalize f_{obj} and f_{subj} to range $[0, 1]$

2. Final Pre-Assessment Score

$$f_{pre} = f_{obj}^{norm} + f_{subj}^{norm}$$

3. Balanced Sampling

- Select equal number of images from each score range
- Avoid bias toward only high- or low-score samples

Annotation Process

- Provided **guidelines** & **anchor images** for rating reference
- Scores: 1-10 (low → high color aesthetics)
- Annotators also gave **confidence score** (0-1)
- Final dataset: images with avg confidence > 0.5
- ~1,500 opinions per image

Experiments – ICAA17K

Metrics:

- SRCC (Spearman Rank Correlation)
- LCC (Linear Correlation)
- Accuracy (positive/negative aesthetics)

Results (ICAA17K)

Model	SRCC	LCC	Accu
Ours *1	0.873	0.890	0.957
Ours *2 (multitask)	0.887	0.901	0.965

- *1: trained only on ICAA17K color scores
- *2: joint prediction of holistic aesthetics (slightly higher performance, longer training)

Experiments – SPAQ

SPAQ Dataset

- Colorfulness score annotation
- Our method: SRCC 0.770, LCC 0.791 (top performance)

Shows ability to generalize to IQA tasks involving color.

Ablation Study

Delegate vs. Swin/DAT Attention

- Delegate outperforms Swin (+5.2% SRCC) & DAT (+4.4% SRCC)
- FLOPs reduced by ~40%

Interest Points & κ

- Too few/many points $\rightarrow$ performance drop
- $\kappa=7$ optimal; higher $\kappa \rightarrow$ more computation, lower $\kappa \rightarrow$ insufficient representation

Visualization

- **GradCAM** shows our model segments image by color regions
- Highest attention typically on **dominant color area**
- Process:
 1. Generate interest points (Net_i)
 2. Adjust offsets (Net_o)
 3. Gate module removes redundancy
 4. Color segmentation module assigns attention per color space

Visualization

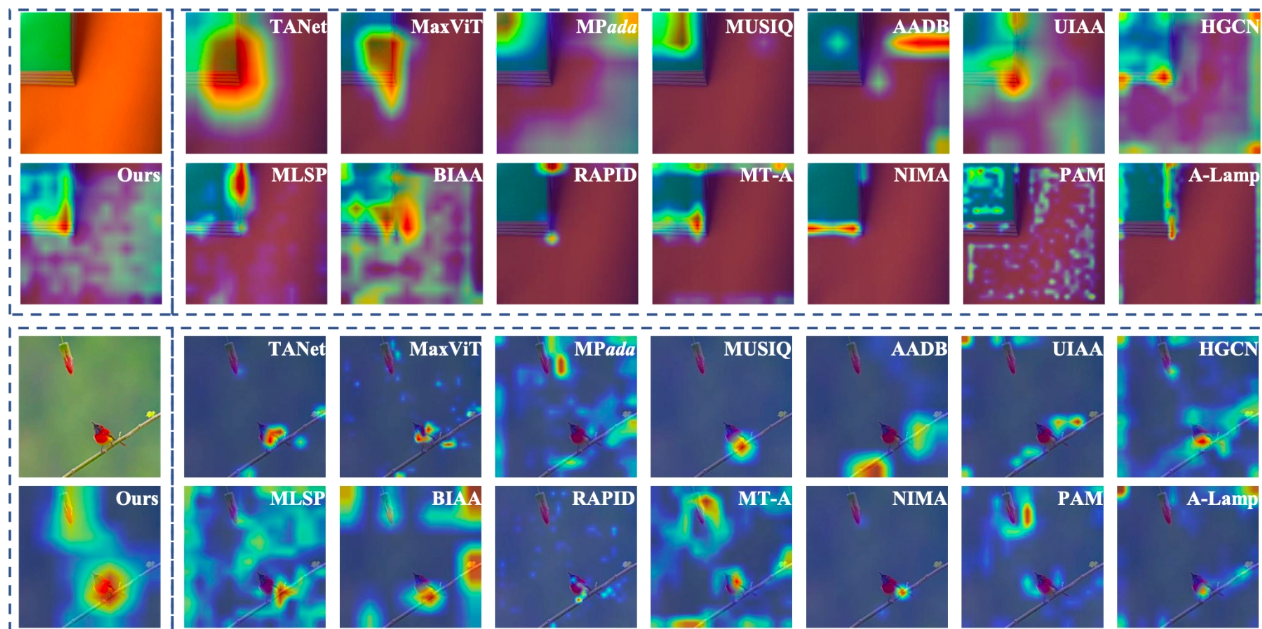


Figure 5. Saliency maps of the 15 models. The attention of our method is segmented into distinct regions based on the color distribution.

Visualization

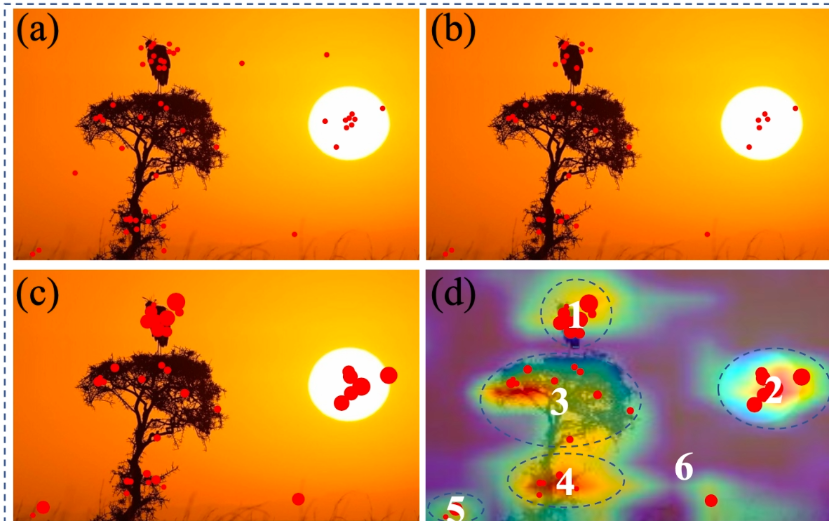


Figure 7. Visualization of our attention mechanism. (a) Sampled interest points adjusted by offsets. (b) Interest points filtered by the gating module. (c) The radius of the circles represents the attention weights at multiple heads, with a larger radius indicating a higher weight. (d) The color space segmentation of each point, where the numbers represent the primary and secondary relationship of the color space, e.g., “1” represents the dominant color.

Conclusion

- First **large-scale dataset** & benchmark dedicated to ICAA
- **Delegate Transformer** effectively models human-like color perception
- Achieves SOTA on both ICAA & color-related IQA tasks
- Future: incorporate more **color prior knowledge**

Thank you